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Second-order velocity structure functions at submesoscales

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ABSTRACT: Observations of ocean surface currents from the JPL Doppler Scatterometer (DopplerScatt) during the S-MODE campaigns reveal unexpectedly shallow second-order velocity structure function (SF) slopes at submesoscale separation scales ($r < 10$ km), deviating from classical turbulence theory and prior modeling results. This discrepancy suggests missing physics in current submesoscale-resolving numerical ocean models or an incomplete interpretation of the DopplerScatt observations. To investigate this, we analyze high-resolution Regional Ocean Modeling System (ROMS) simulations across a range of configurations that isolate the influence of model resolution, season, high-frequency forcings, and surface gravity wave effects on currents. We find that high-frequency motions associated with near-inertial waves reduce the transverse SF amplitude, driving the ratio of longitudinal to transverse SFs close to unity at submesoscales independently of the season. Additionally, the inclusion of wave-current interactions, often omitted in standard submesoscale-resolving models, can produce energetic small-scale motions, leading to broadband shallow structure function slopes. These results reveal a broader mechanism by which shallow structure function slopes can emerge: any process that injects kinetic energy at small scales over a narrow range of wavenumbers will appear broadband in structure function space and produce shallow scalings. Wave effects are one such candidate and offer a plausible interpretation of the DopplerScatt observations under energetic wave conditions. However, under low wave conditions, other processes with similar spectral characteristics are required to account for the observed shallowness. Finally, the relatively large transverse-to-longitudinal SF ratio in DopplerScatt may reflect its lateral averaging over part of an inertial period, a sampling strategy not replicated in models and warranting further study.

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1. Motivations

Submesoscale currents, characterized by their small spatial scales and high-frequency variability [$\mathcal{O}(0.1–10)$ km, from hours to days], play a crucial role in the ocean dynamics (Thomas et al. 2008; Boccaletti et al. 2007; Klein et al. 2011; Fox-Kemper et al. 2011; McWilliams 2016), and in the transfer of energy across scales (Schubert et al. 2020; Srinivasan et al. 2023; Contreras et al. 2023). They modulate biological activity (Mahadevan 2016; Damien et al. 2023) and influence the distribution of tracers in both coastal (Romero et al. 2013; Ohlmann et al. 2017; Dauhajre et al. 2019) and open-ocean environments (Berti et al. 2011; Lumpkin and Elipot 2010; Siegelman et al. 2020; Zhang et al. 2023) with current open questions involving their interaction with near-inertial waves and surface gravity waves.

Ocean eddies distribute wind-forced internal wave energy across scales, contributing to the formation of an internal wave continuum (Yang et al. 2023), while internal waves can modify turbulent energy cascades and drain eddy energy (Barkan et al. 2017, 2021; Delpech et al. 2024; Barkan et al. 2024). The submesoscale-internal wave reaction is more challenging to study and characterize consequent of the overlapping time scales of submesoscale frontal motions and near-inertial internal waves (Thomas et al. 2008; Wang et al. 2023).

Surface gravity waves (SGW) impact ocean currents primarily at submesoscales and smaller scales (Hypolite et al. 2021). The SGW-current interaction can modify three dimensional turbulence in the surface boundary layer (SBL), referred to as Langmuir turbulence, that is formed by tilting of the vorticity anomalies in the SBL into the horizontal through a Stokes drift induced by the SGWs (Craik and Leibovich 1976; Belcher et al. 2012; D′Asaro 2014; Thorpe 2004). At sufficiently high grid resolution, typically in the tens of metres, realistic ocean simulations incorporating these parameterizations begin to partially resolve these Langmuir turbulence processes (Hypolite et al. 2023), evidenced by the emergence of intermittent and heterogeneous roll-cells [$\mathcal{O}(0.1)$ km, lasting a few hours], complementing the insights provided by idealized LES simulations (Sullivan and McWilliams 2019). While these simulations do not fully resolve all scales of Langmuir turbulence, they begin to represent key Langmuir-like dynamics. This partial resolution of small-scale processes improves upon coarser-resolution simulations, yet challenges traditional vertical mixing parameterizations (such as KPP; Large et al. (1994)), and sub-grid scale repre-

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sentation more generally (Wyngaard 2004). Therefore, it is important to validate these models by comparing them with observations.

The NASA Earth Ventures Suborbital-3 S-MODE (Submesoscale Ocean Dynamics Experiment) mission was launched in an attempt to bridge the gap between models and observations by collecting high-resolution data on submesoscale currents (Farrar et al. 2025). This mission successfully deployed a network of in-situ platforms as well as airborne instruments in the Pacific ocean offshore San Francisco, CA, a coast well exposed to ocean swells. The collected datasets, publicly available on the NASA JPL PO.DAAC (https://podaac.jpl.nasa.gov/S-MODE), provide an unprecedented opportunity to increase our understanding of submesoscale dynamics and validate models.

Mean squared quadratic velocity differences over spatial separation scales, typically referred to as second-order velocity structure functions (SF_2 hereinafter) are extensively used within the drifters community to characterize the structure of turbulence in the ocean (Balwada et al. 2016; Poje et al. 2017). Mathematically, SF_2 are closely related to the physical space analogue of the kinetic energy spectrum (i.e. the power spectral density) though the choice of which of the two to compute typically depends on the structure of data.

Classical turbulence theories predict specific power law behaviors for second-order velocity structure functions as a function of separation scale, r, depending on the kind of turbulent flows and regimes; this closely parallels predictions for spectral slopes as a function of wavenumber (Frisch 1995; Pope 2001; Callies and Ferrari 2013). These predictions provide a benchmark against which observed data and model outputs can be compared. However, the data collected by the airborne Doppler Scatterometer instrument (NASA-JPL DopplerScatt) (Rodriguez 2018; Wineteer et al. 2020) during the S-MODE campaigns reveal several deviations at submesoscales from theoretical predictions and standard (defined in Sec. 3.a) realistic ocean model outputs. These deviations suggest that our current interpretation of the nature of the DopplerScatt observations requires refinement, or that the setting of the existing class of submesoscale-resolving numerical ocean models used for comparison is incomplete.

To address these issues, we focus on two key aspects of submesoscale dynamics that reflect the discrepancies between model results and observations: (i) the relative amplitudes of longitudinal and transverse second-order structure functions and (ii) the mechanisms producing shallow SF slopes seen in measurements. Our analysis combines theory with ROMS experiments spanning

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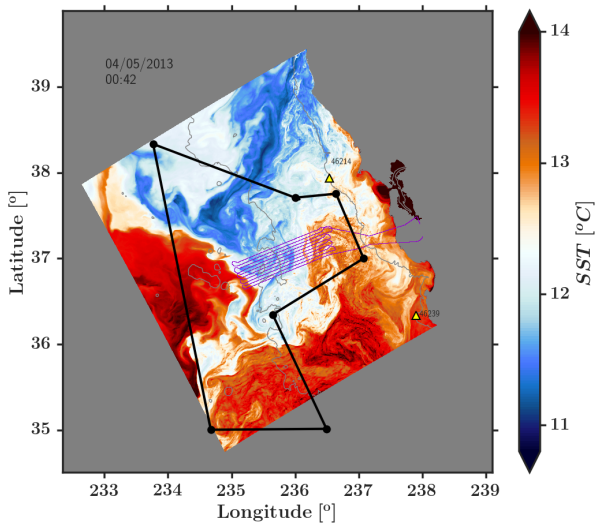

FIG. 1. Snapshot of SST at ocean time $2013/04/05, 00:41:26$ from the SMODE ROMS solution ($\Delta x = 200$m) with the S-MODE operational region (black) and the location of two NOAA NDBC buoys (triangles): Point Reyes (station ID 46214) and Point Sur (station ID 46239). 500m and 4000m isobaths (grey). Track of a B200 flight (violet) performed on the 2023/04/18 (during IOP2) with the DopplerScatt instrument operating.

seasons, resolutions, and forcings (Table 1), using three families of solutions (NISKINe; SMODE; L4 with and without wave effects on currents, WEC) to isolate the roles of high-frequency motions that include near-inertial/internal waves and surface-wave-driven small scales.

Section 2 revisits theoretical expectations for second-order velocity structure functions. Section 3 summarizes the ROMS configurations. Section 4 develops the working expressions for SFs and the small-separation asymptotics relevant to submesoscales. Section 5 tests the impact of high-frequency motions with/without internal-wave forcing, showing that these motions systematically alter the longitudinal-transverse partition, with a decrease in the transverse/longitudinal ratio that is robust across seasons. Section 6 examines WEC and, via a generalized SF formulation that attributes slope contributions to spectral bands, shows how wave-driven small-scale energy produces broadband SF flattening. Section 7 interprets these results for DopplerScatt (S-MODE), and Section 8 synthesizes implications for submesoscale dynamics and for improving numerical models. A secondary objective throughout is a ROMS-DopplerScatt cross-validation, narrowing the model-observation gap at submesoscales.

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2. Deviations between theoretical predictions and DopplerScatt measurements of second-order velocity structure functions

a. Second-order velocity structure functions predictions in the ocean

The Eulerian SF_2 is a statistical metric, dependent on the separation scale r, that provides insights into the energy distribution across flow scales and is linearly related to the two-point correlation function. When comparing two second-order velocity structure functions at a given separation scale, a larger structure function amplitude may indicate a larger energy level or a decreased correlation at that separation scale. Similarly, a steeper structure function slope reflects a faster decorrelation rate with separation scale.

The scale-dependent behavior of turbulence, as characterized by SF, has been identified for a non-divergent isotropic homogeneous ocean (Callies and Ferrari 2013; Soufflet et al. 2016; Bachman et al. 2017), mostly in terms of slopes and the relative amplitude of longitudinal to transverse SFs. Among other things, departures from these laws can then indicate that the flow has significant divergence, inhomogeneity, and anisotropy, which may be typical at submesoscales (Barkan et al. 2019).

1) Spectral vs. structure function slopes

The wavenumber (k) spectra of kinetic energy (KE) with a spectral slope n relates to the SF slope as $k^{-n} \rightarrow r^{n-1}$ for $n \in [1,3]$. Thus, fully developed 3D forward energy cascade turbulence is characterized by a $k^{-5/3}$ law in KE spectra and an $r^{2/3}$ law in SF_2 in (Richardson 1922; Kolmogorov 1941). The inverse energy cascade regime in the ocean, more typical of the mesoscales, is also predicted to have a $k^{-5/3}$ spectral scaling and a $r^{2/3}$ power law in SF_2 , although a k^{-3} spectral slope in the enstrophy range is more commonly observed (Scott and Arbic 2007; Callies and Ferrari 2013). For this forward enstrophy cascade regime in two-dimensional (Kraichnan 1971), or quasi-geostrophic turbulence (Charney 1971), the KE spectra follows a k^{-3} law and an r^2 law in SF_2 . Submesoscale fronts and filaments have sharper velocity gradients than the ones in the enstrophy cascade regime in QG-turbulence and a shallower k^{-2} slope in energy and a corresponding r^1 power law in SF_2 is observed in models and observations (Capet et al. 2008; Pearson et al. 2019; Barkan et al. 2019) (see e.g., Fig. 2 purple and blue cases).

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It is important to keep in mind that small separation scales in structure function space do not always directly correspond to high wavenumbers in spectral space. Thus the $k^{-n} \rightarrow r^{n-1}$ for $1 \leq n \leq 3$ correspondence (Frisch 1995; Pope 2001) between the spectrum and structure function, valid strictly in inertial ranges of turbulent flows, breaks down when the slopes are shallower than k^{-1}, even if this shallow spectral slope region manifests over a narrow band of wavenumbers. This can happen, for example, when energy is injected in a narrow band of wavenumbers consequent of an instability process or as a response to external forcing. Relevant examples include Langmuir turbulence, which can concentrate energy in high-wavenumber bands (e.g., Hamlington et al. (2014)), and near-inertial internal waves, which often exhibit spectral peaks near the local Coriolis frequency (e.g., Alford (2003)). In the latter case, although near-inertial motions are generated at a temporal frequency f, their associated horizontal scales (arising from the wave dispersion relation) can imprint on the spatial structure function. In this case of an energy injection over a narrow range of wavenumbers, even at moderate or high k, the resulting structure function can appear broadband and yield shallow slopes over a wide range of separations (see Sec. 6).

2) Relative amplitude of longitudinal and transverse structure functions

The longitudinal SF_2 ($S_{\parallel}$) measures velocity differences along the direction of separation, while the transverse SF_2 ($S_{\perp}$) measures these differences perpendicular to the direction of separation. For KE spectra, the ratio of transverse to longitudinal spectra follows $K_{\perp}/K_{\parallel} = n$ under assumptions of isotropy and negligible divergence (Batchelor 1953; Leith 1971; Charney 1971), suggesting that the $S_{\perp}$ should also have a larger amplitude than $S_{\parallel}$ at submesoscales (e.g. blue lines on Fig. 2b from Pearson et al. (2019)). Consistent with this expectation, in the limit of $r \rightarrow 0$, previous theoretical results predict limiting values $S_{\perp}/S_{\parallel} = 3$ for two-dimensional turbulence (Babiano et al. 1985), and $S_{\perp}/S_{\parallel} = 2$ for the case of homogenous, isotropic turbulence (Pope 2001). However, the relevance of these limiting results remains unclear for oceanic submesoscale turbulence in lieu of not being characterized by either of these turbulent flow paradigms. For this reason, and since at submesoscales and smaller scales horizontal divergence becomes important (Barkan et al. 2019), we revisit the small scale limit in Sec. 4b. At larger scales, it is expected that the velocities are so decorrelated that their projection onto longitudinal and transverse directions no longer matters, and the ratio of their respective SF should be close to unity. Although the magnitude of the transverse

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and longitudinal SFs (or the spectra) can be different at a given r (or k), the power law should be the same if the flow is isotropic.

In addition to geostrophic and submesoscale flows, internal waves, particularly near-inertial waves, also have clear implications for structure function geometry. In the case of pure inertial motions, the horizontal velocity field is rotationally oscillatory, with the divergence and vorticity cycling out of phase: divergence peaks a quarter of an inertial period before vorticity, and returns to the opposite sign another quarter period later. As a result, inertial flows have no persistent anisotropy in direction, and the transverse and longitudinal structure functions are equal on average, i.e., the SF ratio is 1. This is also approximately true for internal waves with frequencies near the local Coriolis frequency f (except near the equator where $f \to 0$), for example diurnal internal waves especially in the mixed layer, where the horizontal kinetic energy (HKE) dominates the energy budget (Alford 2003). Therefore, in flows where near-inertial waves are a leading component, the SF ratio should be expected to approach unity.

b. Observed second-order velocity structure functions by Doppler Scatterometry

The S-MODE program had a pilot campaign and two operations periods (IOPs) in contrasting seasons, fall 2021, 2022 and spring 2023, during which the DopplerScatt repeatedly measured the ocean surface velocity over multiple parallel lines ~125 km in length, separated by ~4 km, to cumulatively image an area roughly 100 km×40 km in area over a period of ~4 hours at submesoscale resolution (~ 200m – 1 km effectively) (Rodriguez 2018; Wineteer et al. 2020).

Fig. 2a highlights the SF_2 ensemble averaged over the course of IOP-2 observations (spring 2023) consisting of 18 realizations. At the largest available scales, the longitudinal, transverse and total SF are of similar amplitude. Then, for decreasing separation scales, the $S_\perp$ is found to be larger at most scales than the $S_\parallel$. Interestingly, the slope of SF_2 are found to be shallow. The ensemble averaged SF_2 computed over the course of IOP-1 behaves similarly and is not shown. For comparison, we compute SF_2 using a ROMS solution (labelled SMODE; no surface wave; $\Delta x \sim 200$m, see Tab. 1 in the next section for configuration details) and find them systematically steeper, close to r at submesoscales. This behavior is persistent during the entire simulated month.

These model-data comparisons lead us to two key questions that are central to this work:

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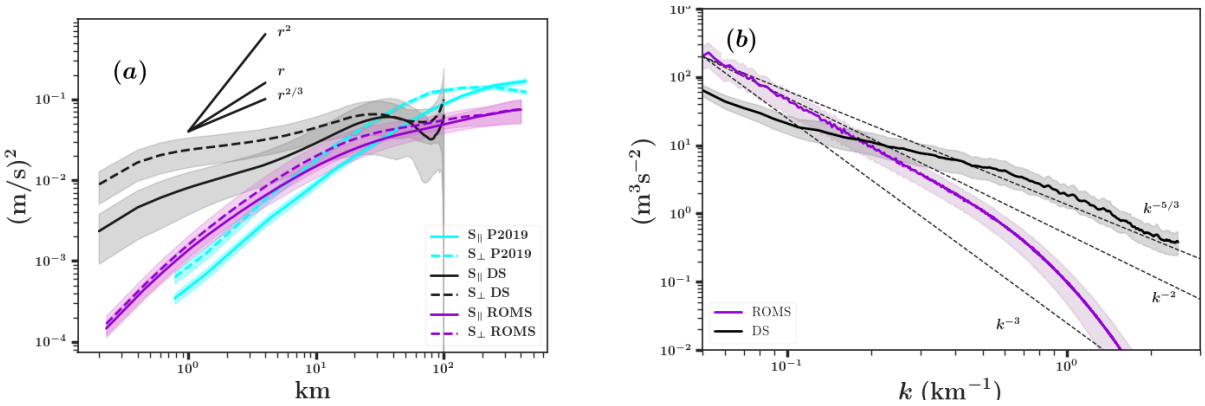

FIG. 2. a: Second-order velocity structure functions (SF) from the DopplerScatt instrument (black) ensemble averaged over the deployment IOP2 (April 2023; statistics are taken over the 200 lines (each 100km) of each day, 18 total for this deployment) in a region offshore San Francisco, CA, as a function of separation scale (km), with averaged daily standard deviations (shade) for the longitudinal (solid) and the transverse (dashed) components of the velocity field. Ensemble azimuthally averaged second-order SF from the SMODE solution (purple) with transverse SF $S_\perp$ (dashed) and longitudinal SF $S_\parallel$ (solid) with standard deviations (from the ensemble average over the 30 days of April 2013). Azimuthally averaged second-order SF from a ROMS smooth 500m resolution solution (blue) with transverse SF $S_\perp$ (dashed) and longitudinal SF $S_\parallel$ (solid), which is from a subset of a Gulf of Mexico (GoM) simulation used in Pearson et al. (2019) (P2019) (also represented in Fig. 5) including lower and upper bounds of confidence interval (shade). b: Mean total kinetic energy spectra with standard deviations (shade) as a function of the spatial wavenumber from the SMODE solution (purple; mean over 30 days of April 2013) and from the DopplerScatt instrument (black) ensemble averaged over the deployment IOP2 (April 2023; statistics are taken over the 200 lines (each 100km) of each day, 18 total for this deployment)

- Why are the amplitude of the longitudinal and transverse structure functions so similar in the SMODE ROMS solution i.e. the ratio $S_\perp/S_\parallel \simeq 1$? (Q1)
- Why is the slope of the structure functions shallower in the DopplerScatt observational dataset than in the SMODE ROMS solution (and shallower than any theoretical expectations, i.e. shallower than $r^{2/3}$)? (Q2)

While both ROMS simulations and DopplerScatt data are structurally compatible with kinetic energy (KE) spectral analysis, we present here both KE energy spectra and second-order structure functions (SFs), and show their consistent behaviors. When examining the KE spectra presented

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in Fig. 2b, DopplerScatt observations show elevated levels of kinetic energy at high wavenumbers, which is related to Q2 as elucidated in Sec. 6. The elevated KE at small scales is related to the flattening of SFs.

The consistency between KE spectra and SFs suggests to use either one of them in an analysis but we find it useful to demonstrate this while documenting SFs behavior in ROMS in different settings. Since SFs tend to be underutilized in the analysis of numerical ocean model outputs, particularly at submesoscales, because they are more costing (unless using spectral operators as we do here, see Sec. 4a) while they are a standard diagnostic for Lagrangian drifter and glider analyses and thus offer a unifying framework for comparing model outputs with observational datasets such as S-MODE, NISKINe, or the ongoing Island Arc Turbulent Eddy Regional Exchange (ARCTERX, project of the Office of Naval Research). Moreover, SFs can be smoother (Pearson et al. 2019) and more robust than spectral estimates under limited spatial sampling, especially when data coverage is uneven or anisotropic (Lumpkin and Elipot 2010).

3. ROMS model configurations

a. Standard run without surface–gravity–wave effects (SMODE, Northern California)

The SMODE ROMS nest ($\Delta x = 200\,\mathrm{m}$; April conditions) is embedded in a Pacific → regional nesting chain and is forced by its parent at 30-min cadence (Table 1). Although not contemporaneous with the S-MODE campaign, matching the seasonal forcing permits statistical comparison with DopplerScatt. The nesting hierarchy is: PACHUG at $\Delta x = 6\,\mathrm{km}$ (100 vertical levels with boundary-layer stretching)[1], NEPAC at $\Delta x = 2\,\mathrm{km}$, WESTC at $\Delta x = 600\,\mathrm{m}$, and SMODE at $\Delta x = 200\,\mathrm{m}$. The parent solutions are forced at the surface by hourly ERA5 (Hersbach et al. 2020), laterally by GLORYS (Lellouche et al. 2018), and include barotropic tides from TPXO9 (Egbert and Erofeeva 2002), enabling baroclinic tides and a realistic internal-wave spectrum to propagate into the nests. River freshwater fluxes enter at the NEPAC stage (Wang et al. 2021; Romero et al. 2016). We treat SMODE as our *standard* configuration: realistic atmosphere, tides, and high-frequency (HF) variability, but *no explicit wave–current interaction* (no WEC).

Our working hypothesis is that HF motions (near-inertial and internal waves) tend to drive longitudinal–transverse partition toward equipartition at submesoscales ($S_\perp/S_\parallel \to 1$). To test this

[1]Configuration details follow McWilliams et al. (2024); Damien et al. (2025).

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TABLE 1. Configuration summary for the three ROMS families used in this study. Each row is a unique run; values are repeated explicitly (no ditto marks). Δx is horizontal grid spacing; "Levels" is the number of vertical z-levels (with BL stretching where noted). "Atm. forcing" indicates the temporal cadence of the atmospheric forcing fields. "Lateral" lists the oceanic boundary condition product; "Tides" include barotropic tidal forcing and baroclinic tides from the boundaries (yes/no). "Waves" indicates whether wave–current interaction (WEC) is included via the wave-averaged vortex force and whether nonhydrostatic vertical momentum is used (NH).

Family	Run (domain)	Δx	Levels	Period	Atm. forcing	Lateral BC	Tides	Waves	Dynamics
NISKINe (North Atlantic; HF vs smooth twins; Barkan et al. 2021; Srinivasan et al. 2023)									
NISKINe	basin/regional (HF)	2 km	100	Feb 2022	hourly	6km run	yes	no WEC	hydrostatic
NISKINe	basin/regional (SMOOTH)	2 km	100	Feb 2022	daily	6km run	no	no WEC	hydrostatic
NISKINe	nested subregion (HF)	500 m	150	Feb 2022	hourly	2km run	yes	no WEC	hydrostatic
NISKINe	nested subregion (SMOOTH)	500 m	150	Feb 2022	daily	2km run	no	no WEC	hydrostatic
SMODE (Northern California; "standard" no-WEC family									
SMODE	PACHUG (Pacific)	6 km	100	Sep 2012–Sep 2013	hourly	GLORYS	yes	no WEC	hydrostatic
SMODE	NEPAC (NE Pacific)	2 km	100	Sep 2012–Sep 2013	hourly	PACHUG	yes	no WEC	hydrostatic
SMODE	WESTC (regional)	600 m	100	Mar–Apr 2013	hourly	NEPAC	yes	no WEC	hydrostatic
SMODE	SMODE (N. Cal)	200 m	100	Apr 2013	hourly	WESTC	yes	no WEC	hydrostatic
Central California L4 (wave–current interaction experiments; Hypolite et al. 2021, 2023)									
L4 chain	L0 (USWC)	4 km	40	Dec 2006–Aug 2007	hourly	6km run	yes	no WEC	hydrostatic
L4 chain	L1 (USWC)	1 km	40	Dec 2006–Aug 2007	hourly	L0	yes	no WEC	hydrostatic
L4 chain	L2 (Central CA)	270 m	50	Dec 2006–Aug 2007	hourly	L1	yes	WEC(WW3)	hydrostatic
L4 chain	L3 (Central CA)	100 m	50	Dec 2006–Aug 2007	hourly	L2	yes	WEC(WW3)	hydrostatic
L4	L4–NO–WEC (open ocean)	30 m	128	Dec 2006 (2 events)	hourly	L3	yes	no WEC	hydrostatic
L4	L4–WEC (open ocean)	30 m	128	Dec 2006 (2 events)	hourly	L3	yes	WEC(WW3)	hydrostatic
L4	L4–WEC–NH (open ocean)	30 m	128	Dec 2006 (2 events)	hourly	L3	yes	WEC(WW3)	NH

Legend — WEC: wave–current interaction via vortex-force formalism (McWilliams et al. 2004; Uchiyama et al. 2010) extended to broadband seas (Romero et al. 2021); WW3: WaveWatch III wave hindcasts (Tolman et al. 2009). Nonhydrostatic option marked NH. Additional forcing products and nesting details appear in SI §S1.

mechanism independently of regional specifics, we use the NISKINe North Atlantic solutions (Barkan et al. 2021), available at $\Delta x = 2\,\mathrm{km}$ and $\Delta x = 500\,\mathrm{m}$, in otherwise comparable pairs *with* and *without* explicit HF forcing ("HF" vs "smooth"; Sec. 5). Grids, forcings, and outputs for SMODE and NISKINe are summarized in Table 1.

b. Runs with surface–gravity–wave effects (central California L4)

To assess whether wave–current interaction (WEC) can generate *shallow* second-order SF slopes (shallower than $r^{2/3}$) by energizing small scales, we analyze a central-California hierarchy culminating in an L4 open-ocean nest at $\Delta x = 30\,\mathrm{m}$ (Table 1). The chain is L0 at $\Delta x = 4\,\mathrm{km}$ and L1 at $\Delta x = 1\,\mathrm{km}$ along the U.S. West Coast (with tides passed to all children), then two submesoscale-

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resolving nests L2 at $\Delta x = 270\,\mathrm{m}$ and L3 at $\Delta x = 100\,\mathrm{m}$, and finally L4 at $\Delta x = 30\,\mathrm{m}$ offshore Point Conception. Vertical mixing uses KPP (Large et al. 1994). Atmospheric fields (precipitation, radiation, winds, temperature, humidity) are from an NCEP NARR-driven WRF configuration (Michalakes et al. 1998) downscaled and applied hourly. WEC is implemented via the wave-averaged vortex-force formalism (McWilliams et al. 2004; Uchiyama et al. 2010) extended to broadband seas (Romero et al. 2021), and is driven by co-located WW3 simulations (Tolman et al. 2009) (L2–L3) interpolated onto L4. Implementation and validation details follow Dauhajre et al. (2017); Hypolite et al. (2021, 2023).

We examine three L4 variants of increasing realism within the same domain and external forcing: (i) hydrostatic, no WEC (L4-NO-WEC); (ii) hydrostatic with WEC (L4-WEC); and (iii) non-hydrostatic with WEC (L4-WEC-NH). Under energetic wave conditions, the WEC cases develop Langmuir-like roll structures ("Langmuir-permitting," i.e., partially resolved circulations with parameterized mixing), which concentrate KE over a narrow wavenumber band and, via our generalized SF-spectrum link, produce *broadband* SF flattening.

Hypolite et al. (2023) identifies two events in L4 characterized by strong wave forcing ($H_s > 4.2$ m) and low turbulent Langmuir number ($La_t < 0.3$; Romero et al. 2021), both exhibiting Langmuir circulation (LC)-like features. Event 1 (12/10/06–12/15/06) occurs under mild winds and is driven by a long-period swell (~2m amplitude), leading to spatially evolving LC-like turbulent structures near the surface that vertically span the entire surface mixed layer over large portions of the domain. Event 2 (12/27/06–12/29/06), associated with high wind speeds ($> 15\mathrm{m.s}^{-1}$) and large, short-period waves (~3m amplitude), generates LC-like structures throughout the domain. These are identified at the surface by narrow, elongated patterns of enhanced vorticity, divergence, and vertical velocity resembling submesoscale filaments, but they are thinner, less skewed, and lack a corresponding surface density gradient. At 30m depth, differences between hydrostatic and nonhydrostatic simulations are minor, suggesting that nonhydrostatic effects are confined to smaller scales. Near the surface, nonhydrostatic vertical velocities are slightly reduced compared to the hydrostatic case, resulting in moderately wider but still energetic LC-like structures, which dominate the kinetic energy spectrum over a narrow band and overwhelm submesoscale motions during Event 2.

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4. Expressions and behaviors of second-order velocity structure functions

a. Definitions

Consider an oceanic surface velocity field $\mathbf{u} = (u, v)$ in the Cartesian coordinate system (x, y), the definition of the Eulerian second-order velocity structure function (SF noted S_2) is

$$S_2(\mathbf{r}) = \langle ||\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})||^2 \rangle_{\mathbf{x}} \,, \tag{1}$$

where $||.||$ is the vector amplitude and $\langle \cdot \rangle_{\mathbf{x}}$ is, in the context of the current work, a spatial (specifically, over $\mathbf{x}$) and time average. It is also customary to compute two structure function variants, the longitudinal SF_2

$$S_{\parallel}(\mathbf{r}) = \frac{1}{r^2} \langle ([\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})] \cdot \mathbf{r})^2 \rangle_{\mathbf{x}} \,, \tag{2}$$

and the transverse SF_2

$$S_{\perp}(\mathbf{r}) = \frac{1}{r^2} \langle ([\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})] \times \mathbf{r})^2 \rangle_{\mathbf{x}} \,. \tag{3}$$

The structure function can to be shown to be linearly related to the two point correlation function as

$$S_2(\mathbf{r}) = \langle ||\mathbf{u}(\mathbf{x})||^2 \rangle_{\mathbf{x}} + \langle ||\mathbf{u}(\mathbf{x}+\mathbf{r})||^2 \rangle_{\mathbf{x}} - 2\langle \mathbf{u}(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}} \tag{4}$$

$$\approx 2\langle ||\mathbf{u}(\mathbf{x})||^2 \rangle_{\mathbf{x}} \, [1 - C(\mathbf{r})] \,, \tag{5}$$

where we assume

$$\langle ||\mathbf{u}(\mathbf{x})||^2 \rangle_{\mathbf{x}} \approx \langle ||\mathbf{u}(\mathbf{x}+\mathbf{r})||^2 \rangle_{\mathbf{x}} \,. \tag{6}$$

This means that the spatial average of the kinetic energy is invariant under translation by $\mathbf{r}$. Although each displacement vector $\mathbf{r}$ samples the field over a differently shifted domain, we assume that the average kinetic energy is uniform across the domain, consistent with local homogeneity. This justifies using a single-domain average to estimate the structure function for all $\mathbf{r}$. Also used in

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Eq.(5) is the normalized correlation function

$$C(\mathbf{r}) = \frac{\langle \mathbf{u}(\mathbf{x}) \cdot \mathbf{u}(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}}}{\langle ||\mathbf{u}(\mathbf{x})||^2 \rangle_{\mathbf{x}}} \,. \tag{7}$$

The expressions for $S_{\parallel}(\mathbf{r})$ and $S_{\perp}(\mathbf{r})$ are substantially more complex than Eq. (5) so we relegate their derivation to Appendix A, with their final forms presented in Eqs.(A3) and (A4). However, the azimuthally averaged form of these functions, obtained by writing $\mathbf{r} \equiv r(\cos\theta, \sin\theta)$ and averaging over θ, i.e.

$$\{S_{\parallel}(r), S_{\perp}(r)\} = \frac{1}{2\pi} \int_0^{2\pi} \{S_{\parallel}(\mathbf{r}), S_{\perp}(\mathbf{r})\} \, d\theta \tag{8}$$

have simple expressions, mirroring the form of Eq. (5) as

$$S_{\perp}(r) = 2\langle ||\mathbf{u}(x)||^2 \rangle_{\mathbf{x}} (1 - C_{\perp}(r)) \tag{9}$$

$$S_{\parallel}(r) = 2\langle ||\mathbf{u}(x)||^2 \rangle_{\mathbf{x}} \left(1 - C_{\parallel}(r)\right) , \tag{10}$$

where

$$C_{\perp}(r) = \frac{1}{\pi} \int_0^{2\pi} \left[C_{uu}(\mathbf{r}) \cos^2\theta + C_{vv}(\mathbf{r}) \sin^2\theta + C_{uv}(\mathbf{r}) \sin 2\theta \right] d\theta \tag{11}$$

$$C_{\parallel}(r) = \frac{1}{\pi} \int_0^{2\pi} \left[C_{uu}(\mathbf{r}) \sin^2\theta + C_{vv}(\mathbf{r}) \cos^2\theta - C_{uv}(\mathbf{r}) \sin 2\theta \right] d\theta \,. \tag{12}$$

Here the normalized correlation functions are defined as

$$C_{uu}(\mathbf{r}) = \frac{\langle u(\mathbf{x}) u(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}}}{\langle ||\mathbf{u}(x)||^2 \rangle_{\mathbf{x}}} , \; C_{vv}(\mathbf{r}) = \frac{\langle v(\mathbf{x}) v(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}}}{\langle ||\mathbf{u}(x)||^2 \rangle_{\mathbf{x}}} , \tag{13}$$

$$C_{uv}(\mathbf{r}) = \frac{\langle u(\mathbf{x}) v(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}} + \langle v(\mathbf{x}) u(\mathbf{x}+\mathbf{r}) \rangle_{\mathbf{x}}}{2 \langle ||\mathbf{u}(x)||^2 \rangle_{\mathbf{x}}} , \tag{14}$$

noting that $C(\mathbf{r}) = (C_{uu}(\mathbf{r}) + C_{vv}(\mathbf{r}))$. Visual inspection of eqs. (11) and (12) trivially leads to the identity, $C(\mathbf{r}) = \frac{1}{2}(C_{\parallel}(\mathbf{r}) + C_{\perp}(\mathbf{r}))$. Substituting this result in eqs. (5), (9) and (10), we get the standard result $S(\mathbf{r}) = \frac{1}{2}(S_{\parallel}(\mathbf{r}) + S_{\perp}(\mathbf{r}))$. Eqs. (5), (9) and (10) are essential for the efficient computation of these structure functions in large grids typical of high resolution ROMS output, as discussed below.

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(i) Computing structure functions in ROMS data The direct, pair-wise calculation of structure functions on ROMS model domains requires $O(N^2)$ computations, where N is the total number of grid points (typically $N \sim 10^6 - 10^7$), which makes this prohibitively expensive. To overcome this, we employ a more efficient spectral approach. First, we utilize the linear relationship between the structure and correlation functions, as established in eqs. (5), (10), (9), (11), and (12). The correlation functions are then computed efficiently using the Cross-Correlation Theorem:

$$\langle u(\mathbf{x})\, v(\mathbf{x}+\mathbf{r})\rangle_{\mathbf{x}} = \int \hat{u}(\mathbf{k})\hat{v}^*(\mathbf{k})e^{-i\mathbf{k}\cdot\mathbf{r}}d\mathbf{k}\,, \tag{15}$$

where $\hat{u}(\mathbf{k})$ and $\hat{v}(\mathbf{k})$ are the Fourier transforms of $u(\mathbf{x})$ and $v(\mathbf{x})$. Note that this is a generalization of the standard Wiener-Khinchin theorem, which applies when $u \equiv v$. The discrete implementation of this theorem using the Fast Fourier Transform (FFT) reduces the computational complexity to $O(N \log N)$, a vast increase in speed on typical ROMS grids.

A crucial property of this method is that its discrete implementation yields the exact two-point correlation function for finite, non-periodic data, as is the case for oceanic fields, without explicitly requiring any data windowing to enforce periodicity. This is achieved provided the inputs are padded with zeros to at least twice the size of the input data along each dimension before performing the FFT. This is a standard result in digital signal processing, which we simply extend and apply from its common use for 1-dimensional time series to our two-dimensional spatial fields. We outline a proof of this identity in SI. A surprising consequence of this result is that, unlike the computation of spectra which requires some method for enforcing periodicity at the boundaries like windowing, computing structure functions through (15) is both discretely exact and efficient to compute.

This method was tested explicitly by computing the structure functions using the direct approach and comparing with those obtained from the FFT approach on a small 256×256 subdomain of the 30 m L4 simulation. Errors less than 10^{-7}% were recorded, while the spectral approach was also found to be about 100 times faster for this case than the direct approach. This efficient computation of two-point correlation functions allows us to compute all the structure functions efficiently across large ROMS grids. We further parallelize the analysis across the time dimension using the Ray multiprocessing library for additional speedup.

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(ii) Computing wave number KE spectra in ROMS data We compute total wavenumber KE spectra $E(k)$ by tapering the horizontal velocity with a Hamming window and applying a 2D FFT. The total second-order structure function, defined as the average of longitudinal and transverse components, is related to $E(k)$ as given in Eq. (20).

b. Asymptotic behavior of SF at small separation scale r - Analytical expression

Following Babiano et al. (1985), B85 hereinafter, we expand the second-order velocity structure function using a Taylor expansion in powers of the separation scale r and explicitly compute the lowest order terms in the expansion (see computation details in Appendix B). We make no assumptions on the structure or complexity of the turbulent oceanic surface velocity field except a certain degree of smoothness required for the Taylor expansions. For subsequent results, beyond the $\mathbf{x}$ spatial average in Eq. (1) through (14), we also perform an azimuthal average in $\mathbf{r}$-space i.e., if $\mathbf{r} \equiv r(\cos\theta, \sin\theta)$, then we average over θ so all the structure functions are only function of the magnitude of the radial coordinate r.

In the $r \to 0$ limit, we find the approximation

$$S_2(r) \simeq \frac{r^2}{8}\langle \zeta^2 + \delta^2 + \alpha^2 \rangle_{\mathbf{x}} \,, \tag{16}$$

where $\zeta = v_x - u_y$, $\delta = u_x + v_y$ and $\alpha^2 = (v_y - u_x)^2 + (v_x + u_y)^2$ are the vorticity, divergence and square of the strain magnitude, respectively. The next order term in Eq. (16) is $O(r^4)$ and given in Appendix B.

Similarly, the lowest order approximations to the longitudinal and transverse structure functions are found to be

$$S_{\parallel}(r) \simeq \frac{r^2}{4}\langle \delta^2 + \frac{\alpha^2}{2} \rangle_{\mathbf{x}} \,, \tag{17}$$

$$S_{\perp}(r) \simeq \frac{r^2}{4}\langle \zeta^2 + \frac{\alpha^2}{2} \rangle_{\mathbf{x}} \,, \tag{18}$$

with the next order terms of both functions also being $O(r^4)$. The dependence of $S_{\parallel}(r)$ on divergence but not vorticity and the opposite for $S_{\perp}(r)$ is expected intuitively: at small separations velocity variations along the axis of separation should be proportional to the δ while transverse

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velocity variance should reflect ζ. However, the identical dependence of both terms on strain is less obvious. To see why, note that the canonical strain field consists of divergence along one direction and an equal magnitude of convergence in the transverse direction; thus both transverse and longitudinal directions should have the same strain dependence.

At first order, these two expressions verify $S(r) = 1/2\left(S_\perp(r) + S_\parallel(r)\right)$ and the ratio between the two is

$$\frac{S_\perp}{S_\parallel} \simeq \frac{\langle \zeta^2 + \frac{\alpha^2}{2} \rangle_{\mathbf{x}}}{\langle \delta^2 + \frac{\alpha^2}{2} \rangle_{\mathbf{x}}} \,. \tag{19}$$

For the special case of homogeneous 2D turbulence on a doubly periodic plane, $\delta = 0$ identically, and it can be shown that $\langle \zeta^2 \rangle_x = \langle \alpha^2 \rangle_x$, thus recovering the results of B85 including $S_\perp(r)/S_\parallel(r) = 3$.

Given the minimal smoothness assumptions required to derive these relations, these scalings should formally be observed in the $r \to 0$ limit for general oceanographic flows, although it is not obvious to estimate at what scales these will be observed. Interestingly, as demonstrated by B85, for the special case of two-dimensional turbulence, the entire enstrophy cascade regime with a $\sim k^{-3}$ regime has a corresponding r^2 power law in SF_2 and not just in the limit of $r \to 0$.

Here, the asymptotic limit in r^2 predicted for small separation scales is unrealized at finite separation scales in both numerical models (see SI for the $r \to 0$ behavior in ROMS) and observations (Lindborg 1999), perhaps because of inaccuracy or noise at the finest scales.

5. Impact of high frequency motions on second-order Eulerian structure functions

a. In the NISKINe solutions

By systematically comparing solutions with and without high-frequency forcing we show that high-frequency and near-inertial motions tend to balance longitudinal and transverse SF and lead to lower $S_\perp/S_\parallel$ ratio at submesoscales.

The NISKINe family is composed of one set of solutions that has high frequency (labelled HF) forcing allowing for near inertial waves and high frequency motions generation, and a second set that does not (labelled smooth) with smooth surface forcing and boundary conditions; see configuration details in Barkan et al. (2021). Each of the HF and smooth solutions is run at two resolutions $\Delta x \simeq 500$m and 2km so there are four NISKINe solutions in total, see Tab. 1.

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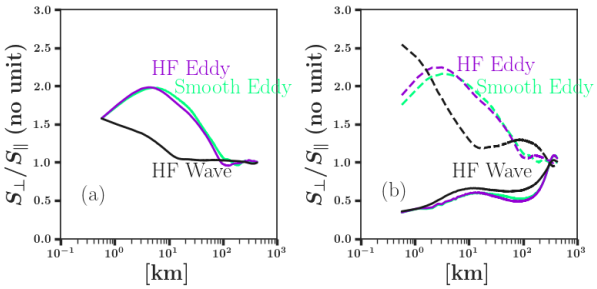

FIG. 3. (a) Azimuthally averaged $S_\perp/S_\parallel$ (no unit) from the 500m resolution NISKINe solution computed in respectively the eddy component of smooth solution (green), the eddy (purple) and wave (black) components of the high-frequency solution. These different components of the flow are further decomposed into their rotational (dashed line) and divergent (solid line) components in b. In both panels, x-axis is the separation scale in km. We perform similar operations on the 2km resolution NISKINe solution and test a different filtering approach and find consistent results (in the supplementary information).

Comparing second-order Eulerian structure functions in these four solutions, at a similar time period, allows to assess the impact of near inertial waves and high frequency motions, along with the effect of resolution. To further pin the mechanism at play to near inertial waves and high frequency motions themselves we separate each solution into eddy and wave components, with near inertial waves and high frequency motions represented by the wave component.

1) Separating eddy and wave components using a time filter

We apply a 20-hour Butterworth filter to the surface velocity fields from the four NISKINe solutions in order to separate the flow into low-pass eddy and high-pass wave components. This Eulerian filtering approach can be affected by Doppler shift effects, as discussed in Jones et al. (2023). For each eddy and wave component, we compute second-order longitudinal and transverse structure functions using a 24-hour average on February 19, 2022. This results in a total of eight cases (four solutions × two components). However, in the two "smooth" solutions, the high-pass (wave) component is weak by construction and therefore not shown. The 20-hour cutoff is selected to approximately separate sub and super-inertial motions in the NISKINe region (~ 68°N, local inertial period is around 13.5 hours).

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We show the results of the time filtering process on the $S_\perp/S_\parallel$ ratio as a function of separation scale in Fig. 3a. The eddy components of both smooth and HF solutions exhibit a similar behavior (purple and green lines). The ratio is close to one at large scales $r > 100$km. Velocities at these scales are decorrelated and therefore are not sensitive to projection onto the longitudinal or transverse directions. The ratio then first increases with decreasing separation scale, and exceeds 2 in the submesoscale separation range $r \in [1-10]$km and before decreasing again moderately at the smallest resolved scales. Modest discrepancies between the 2km and 500m resolution solutions emerge at submesoscales and smaller scales for these eddy components (in the Supplementary Information).

The similarity between the eddy components and the HF wave components is notable at large scales, whereas their divergence becomes apparent at smaller scales. The HF wave solution (black), which is the one that contains high frequency motions, shows a decrease of $S_\perp/S_\parallel$ at submesoscales, compared to the two eddy solutions. This does not seem to be affected by resolution here (in the Supplementary Information).

2) Helmholtz decomposition

We decompose the eddy and wave components in each solution using a Helmholtz decomposition and compute the SF of the rotational and divergent part of the flow in Fig. 3b in each case.

The association between the divergent and rotational components of the flow and the longitudinal and transverse structure functions, respectively, follows from their geometric influence on velocity differences. Divergent flows, by definition, produce stretching or compression of fluid parcels along the direction of separation, thereby enhancing the longitudinal velocity differences ($S_\parallel$). Rotational flows, in contrast, induce swirling motions that result in velocity differences perpendicular to the separation vector, enhancing the transverse structure function ($S_\perp$). Hence, when the divergent component dominates, we expect $S_\parallel > S_\perp$ and $S_\perp/S_\parallel < 1$, while in rotationally dominated flows, $S_\perp > S_\parallel$ and $S_\perp/S_\parallel > 1$. This correspondence makes it common to interpret longitudinal structure functions as more sensitive to divergence, and transverse ones as more indicative of vorticity, and strictly they are related via the Lindborg (2015) relations. The smaller ratio (than in the eddy components) at submesoscales in the HF wave case observed so far in the HF Wave solution, is

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imprinted on the rotational component solely. This effect is more important at higher resolution (shown in the Supplementary Information).

This suggests that high frequency motions mostly impact the rotational part of the flow at submesoscales and reduce their relative $S_\perp$ amplitude. This relative decrease of the rotational transverse energy illustrates the transition in the energy distribution from large separation scales, where mesoscale mostly rotational motions dominate to submesoscale divergent currents domination. This view is consistent with near-inertial waves having similar amplitude vorticity and divergence fields, which intuitively should result in similar energy levels of transverse and longitudinal SF.

The same behavior is observed when using the method described by Wang et al. (2023) to decompose the flow into eddy-wave components (in the Supplementary Information), suggesting that this result is not sensitive to the wave-eddy decomposition method.

b. In the SMODE solution

We have shown that near-inertial waves and high-frequency motions lead to the ratio of $S_\perp/S_\parallel$ to be close to one at submesoscales in the NISKINe solutions. We demonstrate here that it is the reason why we obtain $S_\perp/S_\parallel$ close to one in the SMODE ROMS solution as well. To do so, we use the same time filter as in Sec. 5a and show the resulting SF of the eddy and wave components in Fig. 4 a and b respectively. The eddy component, i.e. the part of the solution without wave or higher frequency motions, has well separated longitudinal and transverse component SF. This is expected for the submesoscale range where vorticity, primarily feeding into the transverse SF, is important and larger than divergence, primarily feeding into the longitudinal SF. By contrast, the wave component, containing all the high-frequency motions including near-inertial waves have again $S_\perp/S_\parallel$ very close to one, illustrating the equipartition of these motions in vorticity and divergence.

c. Weak seasonality of high-frequency motions equipartitioning SFs

Across multiple ROMS configurations, covering different seasons, we evaluate the ratio of transverse to longitudinal second-order velocity structure functions $S_\perp/S_\parallel$ separately for eddy and wave components. We find that the wave associated ratios consistently remain close to one across submesoscale separation scales, indicating that high-frequency motions (contained solely in the

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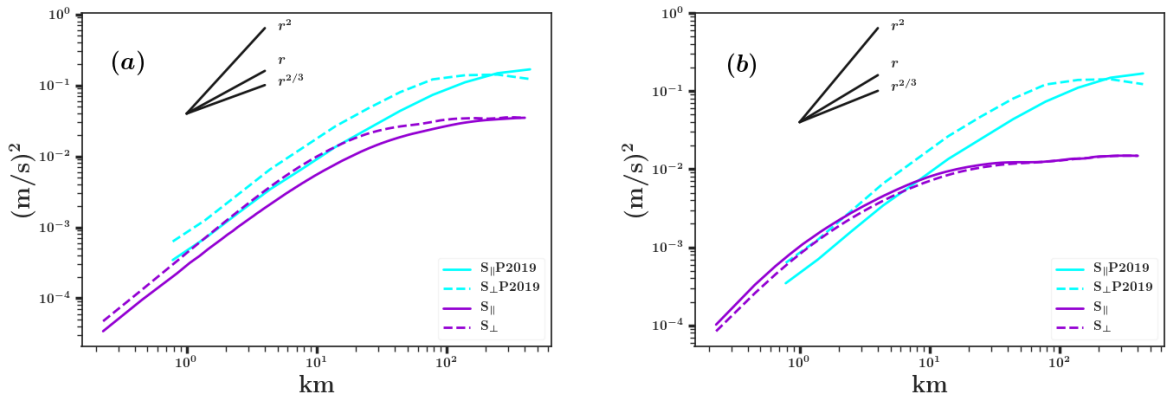

FIG. 4. a: Azimuthally averaged $S_\perp$ (purple dashed line) and $S_\parallel$ (purple solid line) from the eddy part of the flow in the SMODE solution; we overplot the solution shown in Pearson et al. (2019) (cyan lines) for comparison which is a subset of GoM smooth 500m. x-axis is the separation scale in km. Ocean time is $2013/04/05, 00:41$ with $\langle\cdot\rangle$ a 24 hour average. b: Same as in a but for the wave part of the flow.

wave component) equipartition the SFs independently of the time of the year. This behavior is confirmed by a summer simulation (NISKINe HF summer in Tab. 1) that exhibits similar near-unity ratios at most scales (see solid orange lines in Fig. 5a). In contrast, the eddy component exhibits larger and potentially seasonally varying ratios, though resolution constraints limit a definitive attribution. We summarize this finding in the boxplot in Fig. 5b where we show the mean transverse-to-longitudinal structure function ratios for separation scales smaller than 10km, separating wave and eddy components. Wave components consistently yield near-unity ratios (see green box and horizontal dashed line in Fig. 5b), while eddy components (orange box in Fig. 5b) show larger values, suggesting that high-frequency motions equipartition SFs independently of configuration or season.

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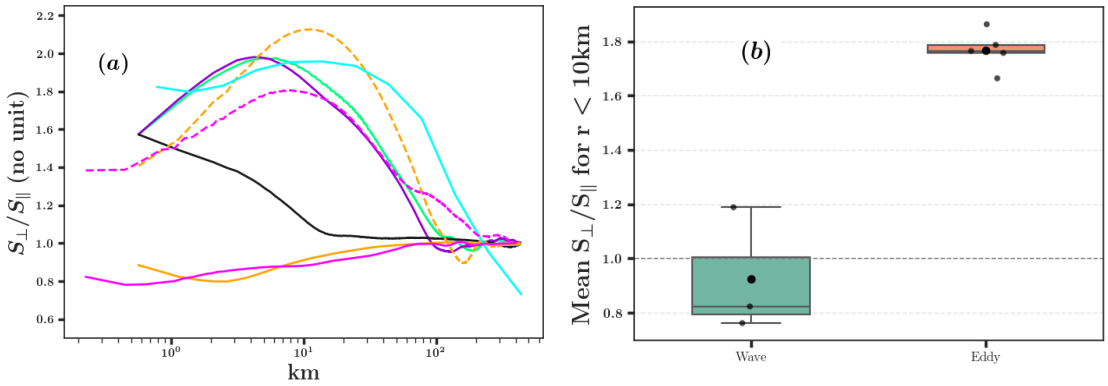

FIG. 5. a: Azimuthally averaged $S_\perp/S_\parallel$ (no unit) from different solutions: the HF NIKSINe eddy and wave components (winter; purple and black respectively), the smooth NIKSINe solution (winter; green), SMODE solution (spring; pink; eddy component in dashed and wave in solid line) and NIKSINe solution during the summer time (summer see Tab 1; orange; eddy component in dashed and wave in solid line), the GoM solution from Pearson et al. (2019) (blue). The NIKSINe winter cases HF and smooth are the same as represented in Fig. 4a. The SMODE case is the same as used in Fig. 5. b: Radial mean $S_\perp/S_\parallel$ (no unit) averaged over submesoscale separation scales ($r < 10$km) across the ROMS simulations. Each point represents a separate case. The mean value is depicted by a larger circle and the InterQuartile Range (IQR) with colored boxes.

6. Impact of waves on second-order Eulerian structure functions

a. Second-order structure functions and kinetic energy spectra

If the flow is isotropic, the one-dimensional total second-order velocity structure function is related to the kinetic energy spectrum of the velocity field via

$$S_2(r) = \int_0^{+\infty} E(k)dk - \int_0^{+\infty} E(k)J_0(kr)dk \,, \tag{20}$$

(Xie and Bühler 2019; Bennett 1984), where J_0 is the zeroth order Bessel function of first kind and the energy spectrum is $\langle \frac{(u^2+v^2)}{2} \rangle_x = \int_0^{+\infty} E(k)dk$, with k the radial wavenumber. This relationship can be derived by combining Eq. (15) (the cross-correlation theorem) with Eq. (5) connecting $S_2(\mathbf{r})$ and the correlation function, with the assumption of statistical isotropy. We provide an explicit derivation of this in SI for a general flow, illustrating the precise role of isotropy. For general

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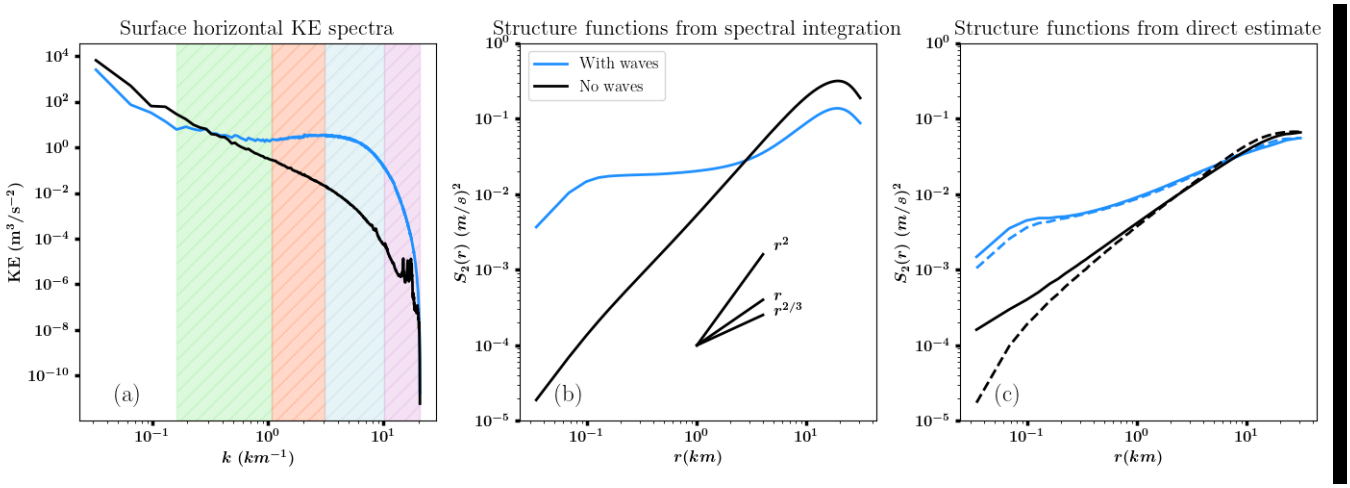

FIG. 6. a: Time average surface horizontal [m^2s^{-2}/(cycles/m)] kinetic energy power spectra during Ev. 1 without wave (black) and Ev. 2 with waves (blue), see Tab. 2 for corresponding ocean time. We highlight the different wavenumber ranges: large scales (green), submesoscales (orange), smaller scales (blue), and dissipation scales (violet). b: Total second-order velocity structure function resulting from the integration of the spectra per Eq. (20) for the same cases as in a. c: Azimuthally averaged second-order velocity structure function resulting from the direct computation per Eq. (10) (solid line) and Eq. (9) (dashed line) for the same cases as in a.

anisotropic flows as is found in the high resolution ROMS solutions, one should not expect Eq. (20) to hold. However, as we demonstrate in the next section, it remains a useful basis for analyzing the structure function.

b. Piecewise spectral integration in ROMS solution

There is observational evidence of wind-wave interaction in the Gulf of Mexico with second-order velocity structure function shallower than $r^{2/3}$ (Chang et al. 2019) which suggests that Langmuir circulations, phenomena operating at small separation scales, could be responsible for the shallow slope. We use the L4 solutions (see Tab. 1), in which Langmuir circulations-like structures start to be resolved, to test if this behavior is consistently observed with ROMS. For these solutions, KE spectra are computed and documented in Hypolite et al. (2023). Under large wave forcing, KE spectra can show a significant enhancement of energy level in the surface mixed layer at small scales (< 200 m) similar to what has been found previously in Langmuir turbulence resolved in LES studies, e.g. Hamlington et al. (2014).

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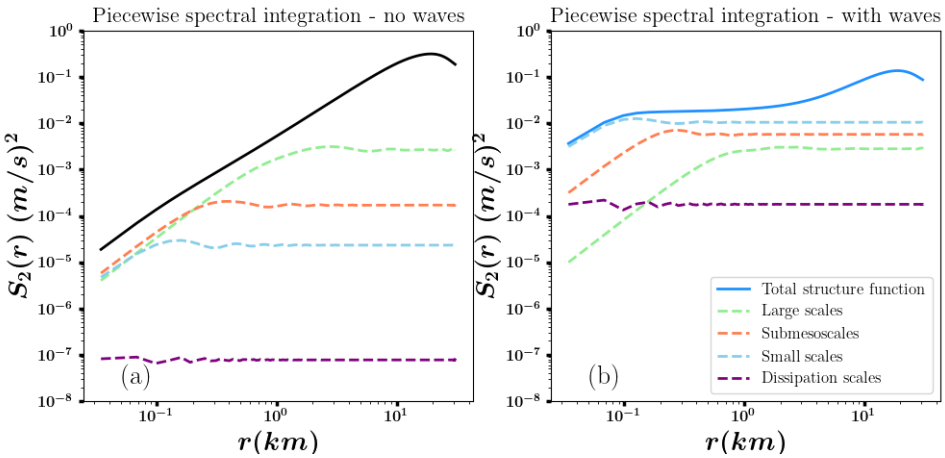

FIG. 7. Second-order velocity structure function resulting from the integration of the spectra per Eq. (20). a: During Ev. 1 without wave. Total second-order velocity structure function (black). b: During Ev. 2 with waves (blue). Dashed lines represent the contribution of each radial wavenumber range as highlited in Fig. 6a with the same color-coding.

We reproduce two cases in Fig. 6a: During Event 1 without wave (see Tab. 1, and Fig. 6a black lines), which contains a highly resolved submesoscale soup, and during Event 2 with WEC (see Tab. 1, and Fig. 6a blue lines), where partially resolved Langmuir circulations dominate the small-scale dynamics.

Using Eq. (20), and assuming isotropy for the moment, we compute the SF from the integration of the KE spectra in Fig. 6b to be compared to the direct SF computation along the longitudinal (solid line) and transverse (dashed line) directions in panel c. Results from Eq. (20) and direct SF computation agree qualitatively in amplitude and slope, but we note that the sum does not match exactly. There are clear quantitative differences, for example at large and small separations. We attribute this to the isotropic assumption made for the derivation of Eq. (20) whereas the direct estimation performs an azimuthal averaging after computing the two-dimensional $S_2(\mathbf{r})$.

The case without waves (black lines in a, b, c) shows expected submesoscale behavior with a slope close to r in SF and a red spectrum in the submesoscale range. The case with WEC (blue lines in a, b, c) departs from this behavior with a blue spectrum in the submesoscales and small scale k range $[10^{-3}, 3\times10^{-3}]$ m^{-1}. The spectrum reaches a local maximum at the length scale $l \simeq 160$ m, which corresponds to the width of the partially resolved Langmuir circulations at this

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resolution (estimated independently by measuring statistically the width of the alternating sign convergent lines at surface).

Similarly to large scales, both simulations' KE spectra agree at dissipation scales. The SF associated with the blue spectrum has a shallower SF than the no-waves case. We arbitrarily define four k ranges (color-coded in Fig. 6a) to highlight the respective contribution of large scales (in green), submesoscales (blue part of the spectra; shown in orange), smaller submesoscales and Langmuir circulation scale (in blue), and dissipation scales (in purple). The results in Fig. 7 highlight the contribution (in dashed lines with the same color coding) of these k ranges from the KE spectrum to the SF (shown in solid lines).

It shows that the peak in KE near 160m is responsible for the flattening of the SF (blue dashed line, Fig. 7b). We display the case without wave for reference, where each k range contributes monotonically (from small to large scales) to the total SF without any flattening. With waves, the largest contributions are from the small scales, and submesoscales, even to a certain degree the dissipation scales. The largest scales contribute the same amount to the total SF in both cases, however, in the presence of waves, small scale contributions are significantly larger. Crucially, this reveals that high wavenumber motions have an important impact even at large separations scales. *Any* process that injects energy at small scales over a narrow range of wavenumbers will appear broadband in structure function space and produce shallow scaling. Wave effects are one such process as exemplified here by partially resolved Langmuir circulations in ROMS producing shallow (shallower than $r^{2/3}$) second-order velocity structure function at submeso-separation scales with WEC.

We observe a transition from steep to shallow structure function (SF) slopes, which correlates with the presence and strength of Langmuir circulations in the solutions. This transition is illustrated in Fig. 8 across five cases. The black and blue lines correspond to the same cases shown in Fig. 6c. In the blue and pink cases, where Langmuir circulations develop across the entire domain, the slopes of the second-order velocity structure functions are shallower than $r^{2/3}$. Conversely, in the black case, which lacks waves, wind, and Langmuir circulations, the SF slopes are steep.

Intermediate cases are represented by green and purple lines: The purple case reflects the L4 NO WEC solution, during periods of high winds (event 2), where sparse wind roll cells are present. The green case represents the L4 WEC solution during periods of large swell and low wind (event

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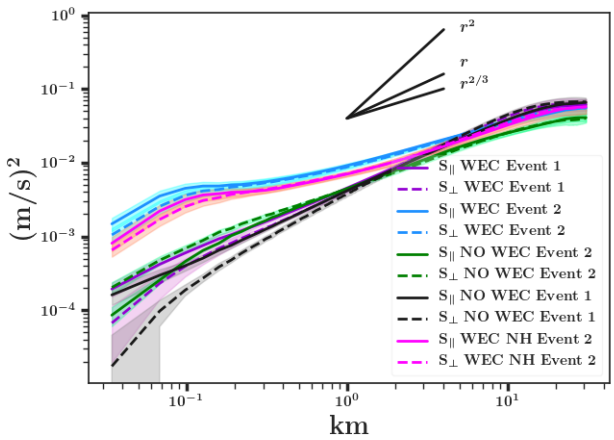

FIG. 8. Azimuthally averaged $S_\perp$ (dashed) and $S_\|$ (solid) from the L4 WEC and NO WEC solutions, with $\langle\cdot\rangle$ a 24hrs average at different times illustrating different Langmuir turbulence development: L4 NO WEC solution with no Langmuir circulation at all (black), ocean time is 2006/11/12 (Event 1), with low wind conditions, same as in Fig. 6, and with waves (L4 WEC) with intermittent Langmuir circulations (purple) at same time of low wind but large swell. At ocean time 2006/28/12, L4 NO WEC at times of high wind (green) (Event 2; sparse wind roll cells are present) with waves (L4 WEC) at times of high wind and large waves (blue) (Langmuir circulations present everywhere at all time, same as in Fig. 6). L4 WEC Non hydrostatic (NH) during same high wind and large waves event (pink). x-axis is the separation scale in km. Shades show the standard deviations from the 24hrs averages.

1), characterized by intermittent partially resolved Langmuir circulations. These intermediate cases exhibit SF slopes that lie between the two extreme scenarios, thereby bridging the gap and corroborating the influence of Langmuir turbulence in driving this transition.

These results remain consistent regardless of whether using the non-hydrostatic solver (pink case) and whether Stokes drift is included in the SF computation. The influence of Stokes drift appears to be secondary here.

7. DopplerScatt measurements interpretation and discussion

a. DopplerScatt second-order structure function slopes and surface waves

Direct comparison between ROMS outputs and observations at small scales is challenging due to resolution limitations and other factors. Nonetheless, analyzing second-order velocity structure

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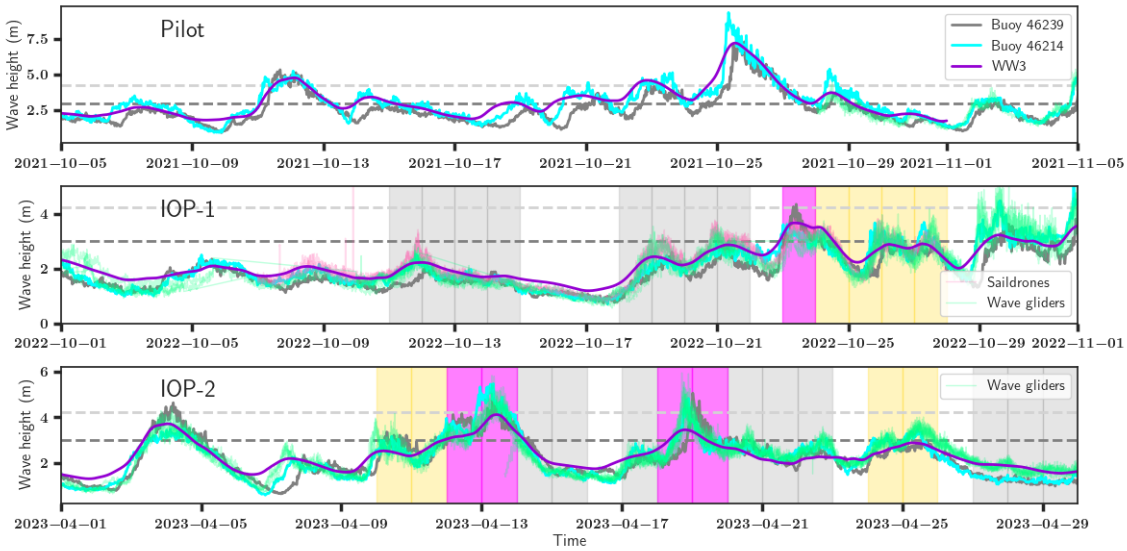

FIG. 9. Timeseries of spatially averaged significant wave height (m) in the operational region of the NASA mission S-MODE produced by WW3 (purple) and as reported by the Point Reyes NOAA buoy (station ID 46214) (blue) and by the Point Sur NOAA buoy (station ID 46239) (grey) during the pilot and the two IOP observational S-MODE campaigns. Significant wave height measured in-situ by saildrones (pink) and wave gliders (green). Days with DopplerScatt operating with a calm sea (i.e. when the significant wave height is below 3m - grey), or when one platform measures it between 3m and 4.2m (yellow), or exceeding 4.2m (magenta).

functions, a statistical measure, can yield valuable insights into the model's accuracy relative to observed surface currents and Langmuir dynamics representation.

We observe that the shape (specifically, the slope and total amplitude) of the blue curves in Fig. 6b,c closely resembles the DopplerScatt observations depicted in Fig. 2. This similarity may indicate that the DopplerScatt captured small-scale current dynamics influenced by surface gravity waves and possibly Langmuir turbulence, manifesting as small oceanic roll-cells.

To explore this possibility further, we present the wave conditions (total significant wave height) during the S-MODE observational campaigns in Fig. 9. We pull the NOAA buoy data from the Point Reyes station buoy (station ID 46214) in cyan from the Point Sur buoy (station ID 46239) in grey. Their locations are shown with yellow triangles in Fig. 1. We also show in purple the spatially averaged total wave height from the Wave Height, Water Depth, and Current Hindcasting (WAVE-WATCH III, WW3) third generation wave model (Tolman et al. 2009) global products at

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half a degree resolution (Cheung 2010) distributed by the Pacific Islands Ocean Observing System (PacIOOS). The spatial average is performed over the operational region (we use the latitude bounds $[34;40]^\circ$ and the longitude bounds $[-128;-120]^\circ$). The WW3 and buoys data are in very good agreement with the significant wave heights measured by wave gliders (Stokes, Kelvin, and WHOI 43) during the pilot, by saildrones (1026,1048,1085, and 1087) and wave gliders (Planck, Stokes, Kelvin, Pascal, Carson, and WHOI 22,32, and 43) and during the IOP1, and by wave gliders (Planck, Stokes, Kelvin, Pascal, Carson, and WHOI 22,32,43, and 1102) during IOP2, using 5min averages. Throughout the pilot and the two IOP campaigns, the average wave height in the operational area often exceeds 3 m, a threshold below which the sea state is generally considered calm (see dark grey dashed horizontal line in Fig. 9). The wave height sometimes surpasses 4.2 m (light grey dashed horizontal line), a condition under which Wave Effects on Currents (WEC) becomes significant, as demonstrated in Hypolite et al. (2021) for 300m and 100m resolutions. Under these large wave conditions, LC-like structures, characterized by greater numbers and finer structures at higher resolutions, are likely to emerge. Note that it is reasonable to expect that the threshold wave height for the emergence of Langmuir turbulence may decrease with increasing resolution in simulations, and therefore in the field, as observed in Hypolite et al. (2023) where higher resolutions revealed a greater sensitivity to lower wave heights.

In Fig. 9, days with DopplerScatt data collection under conditions where the spatially averaged significant wave height (H_s) exceeds 3 m are highlighted in yellow and magenta, while those under low wave conditions ($H_s \leq 3$ m) are highlighted in grey. When considering all yellow and magenta days, we find that the averaged second-order SF exhibit moderately shallower slopes in both directions (compared to the averages over the grey days), as shown in Fig. 10a. Further distinguishing the days with largest waves ($H_s > 4.2$ m), represented with magenta in Fig. 9, reveals even shallower slopes in the averaged second-order SF, as illustrated in Fig. 10b. This suggests a possible influence of Langmuir turbulence, indicating that the DopplerScatt measurements may have captured small-scale current dynamics affected by surface gravity waves on these days.

However, the averaged SF under low wave conditions (black curves in Fig. 10) also display a shallow slope, suggesting that wave-current interactions alone do not fully account for this. Another mechanism is necessary to comprehensively interpret the DopplerScatt dataset at small scales under small wave conditions. This study suggests that such mechanism most likely share the

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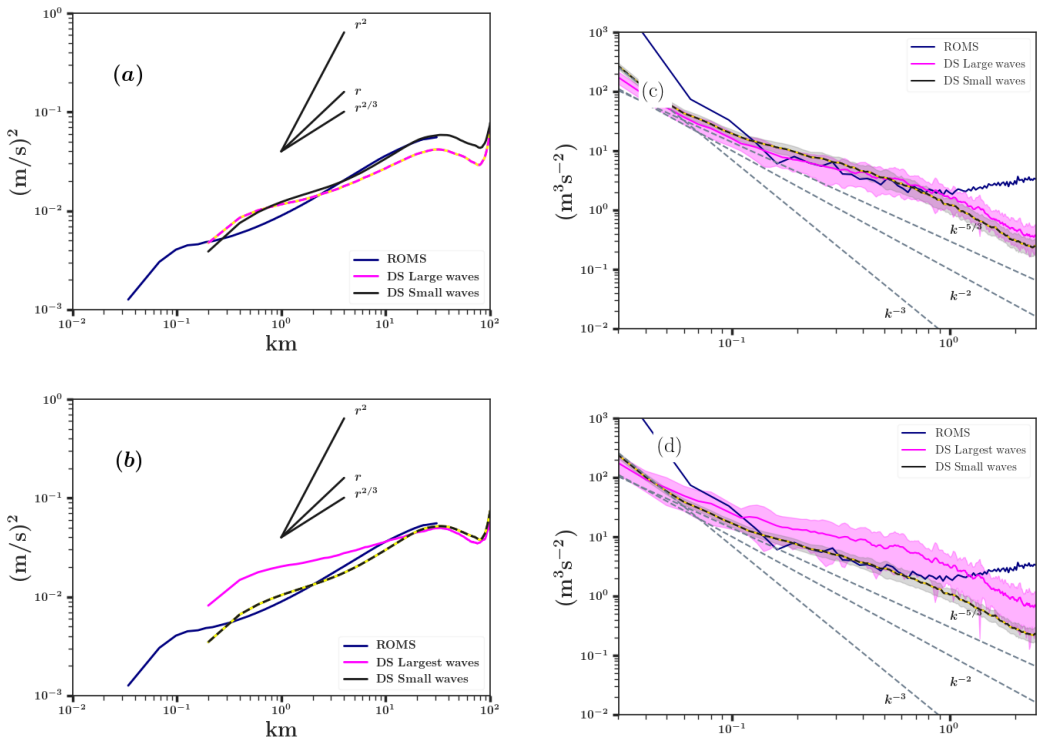

FIG. 10. Total $(1/2(S_\perp + S_\parallel))$ second-order velocity structure functions as a function of separation scale (km) from the DopplerScatt instrument and the L4 WEC ROMS solution (blue), averaged over 24 hours. In ROMS, ocean time is 2006/12/28, with well-developed Langmuir turbulence. (a) DopplerScatt data are ensemble averaged over low (black) and moderate to large (magenta-yellow dashed) wave conditions. (b) DopplerScatt data are ensemble averaged over low to moderate (black-yellow dashed) and largest wave days (magenta) following the color-coding of Fig. 9. Figures a,b with standard deviations is in SI. (c) and (d) are the total kinetic energy spectra using the same color-coding as in (a) and (b) respectively.

spectral properties of wave-current interactions depicted in Sec. 6, namely a narrowband elevated KE spectrum at large k.

For future work, the pilot campaign should also be an interesting period to further test the correlation between wave height and shallow second-order SF as the wave height was consistently large.

In this study, we use significant wave height (H_s) as a practical and conservative proxy for wave influence on upper ocean dynamics, including conditions potentially favorable for Langmuir turbulence. While the Langmuir turbulence number ($\mathrm{La}_t = \sqrt{u*/U_s}$), where $u*$ is the frictional

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velocity and U_s the surface Stoke drift, is often used in theory to predict the strength of Langmuir turbulence (Belcher et al. 2012), its applicability here is limited by the difficulty of accurately estimating the surface Stokes drift U_s. This task requires knowledge of the full 2D wave spectrum, which is not retained during SMODE. Instead, wave measurements are reduced to either 1D spectra or bulk parameters such as H_s, peak period T_p, and peak direction θ_p, leading to systematic underestimation of U_s (Kumar et al. 2017) and thus inflated values of La_t. For this reason, computed La_t values rarely fall below 0.3 during the S-MODE IOPs, and show limited distinguishing power between regimes of potential Langmuir activity. Given these constraints, H_s appears as a more robust and directly observed indicator of wave influence in general, accepting that it is less strict than La_t at filtering Langmuir turbulence conditions. But, previous modeling studies (Hypolite et al. 2021; Romero et al. 2021) (see respectively their Fig. 2 and Fig. 14) show that H_s and La_t identify the same time periods as energetically significant wave events, labelled Event 1 and Event 2 here (tab. E1), both characterized by large wave height $H_s > 4.2$m and $\mathrm{La}_t \leq 0.3$. In this context, H_s permits to identify how energetic surface wave conditions may drive small-scale, narrowband processes that lead to shallow structure function slopes.

b. Error and sampling effects discussion

1) Relative amplitude and inertial waves

If surface gravity wave effects on currents provide a framework for interpreting shallow SF slopes at times, they do not account for the relatively large ratio of transverse to longitudinal SF. However, we demonstrate in Sec. 5 that near-inertial waves and higher-frequency motions can decrease this ratio in the submesoscale range. A possible explanation for the absence of this observation in the DopplerScatt dataset might relate to the sampling lateral averaging technique over half (or 2/3) of an inertial period.

We attempt to replicate this sampling effect by applying a 6-hour averaging window and by dividing the region into four subregions, but we do not observe any significant differences in the resulting SF that would support this hypothesis.

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2) Error analyses

To assess the robustness of the SF analysis, we include error bars in Fig. 2, Fig. 8 and Fig. 10 (also in SI) computed from the variability across hourly realizations in SMODE (15min in L4) and from jackknife resampling in the DopplerScatt (DS) observations (Rodríguez et al. 2018). In the model, SF slopes and amplitudes are stable across 30-day averages in SMODE and within the 24-hour windows used for the L4 simulations, with only minor dispersion near the resolution scale where such uncertainty is expected due to grid discretization. The DS observations exhibit wider error bars, reflecting the larger variability of the observed surface currents, but the slopes remain consistent across scales. The difference in slopes between longitudinal and transverse SFs in DS cannot be explained solely by uncertainty and likely reflects a degree of anisotropy in the flow.

Instrumental noise in DopplerScatt could also contribute to the observed differences in SF slopes or to the elevated transverse-to-longitudinal SF ratio. However, the shape of the SFs in DS differ from what would be expected from spatially uncorrelated noise, which typically introduces a flat scale-independent contribution at small separations. Furthermore, the SFs remain coherent with both the longitudinal and transverse SFs exhibiting scale-dependent power-law behaviors across a wide range of separation distance, which suggests physically coherent variability rather than random measurement error.

8. Summary

In this study, we investigate the analytical and numerical behavior of second-order velocity structure functions, through both idealized theoretical derivations and the analysis of ROMS simulation outputs. We aim to address questions Q1 and Q2 as outlined in the introduction and Sec. 2b.

For Q1, we assess the impact of high-frequency motions and near-inertial waves on second-order velocity structure functions using the NISKINe (Thomas et al. 2020; Barkan et al. 2024) ROMS output dataset. By systematically comparing solutions with and without high-frequency forcing, we observe a decrease in the ratio of transverse to longitudinal components of the second-order velocity structure function from 2.5 to 1, indicating the influence of high-frequency motions at sub-mesoscales. Decomposing the flow into eddy and wave components, the decrease is observed only in the wave component, with the eddy component aligning with wave-less theoretical predictions.

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This pattern is further validated using the SMODE solution. A Helmholtz decomposition reveals that high-frequency motions predominantly affect the rotational component of the flow rather than the divergent component, and we find that the decrease is more pronounced at higher resolutions.

Regarding Q2, we evaluate the influence of surface gravity waves on second-order velocity structure functions using central California ROMS solutions that include or exclude surface gravity waves (Hypolite et al. 2023). We derive a generalized formulation of the second-order velocity structure function to delineate the contributions of different kinetic energy spectra ranges to the slope of the structure function. This method demonstrates that surface gravity waves, through the generation of an energetic small-scale process resembling Langmuir circulations, can flatten the second-order velocity structure function. We observe a transition from steep to shallow SF slopes, correlating with wave height and the presence and intensity of the Langmuir-like circulations, thereby corroborating their role in driving this transition.

During the S-MODE campaigns, our investigation suggests that large wave events, highlighted in magenta in Fig. 9, are associated with shallower second-order velocity structure function (SF) slopes compared to other collection days, as measured by the DopplerScatt system. But, this analysis also makes clear that another mechanism is needed to explain the shallow SF slopes observed by DopplerScatt during low wave days.

Beyond the technical evaluation of second-order velocity structure functions in ROMS and DopplerScatt data, our results have several broader implications for ocean turbulence diagnostics and modeling.

First, by revealing that wave components alone drive $S_\perp/S_\parallel$ toward unity, this work suggests that DopplerScatt, which operates at the surface and captures high-frequency variability, may be particularly sensitive to near-inertial and wave-induced dynamics. Satellite-based Doppler scatterometer (like the proposed ODYSEA), with the ability to cover large areas in a shorter time, may be able to partially mitigate this aliasing effect. Thus future spaceborne implementations may yield SF slopes more representative of the mesoscale-submesoscale energy distribution, rather than the surface expression of transient high-frequency dynamics.

Second, we show that energy injections over a narrow range of large wavenumbers yield broadband flattening of SF slopes. This provides a general physical mechanism that can apply beyond wave forcing. It opens the possibility that other small-scale, energetic processes, such as subme-

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soscale frontal and mixed layer instabilities, could also manifest as shallow SF under low-wave conditions. This broader framework will inform model validation efforts and new parameterization strategies for high-resolution models targeting the surface boundary layer.

Finally, by highlighting where and how current ROMS setups diverge from DopplerScatt measurements, our analysis emphasizes the need for improved coupling between wave models and circulation models in submesoscale-resolving simulations, especially for regions where Langmuir turbulence can develop.

Together, these insights suggest that the structure function framework, when interpreted in light of surface wave and high-frequency dynamics, can serve as a diagnostic bridge between theory, models, and remote sensing observations of ocean turbulence. The findings advocate for enhanced integration of wave-current processes in both observational analysis and numerical modeling to better capture submesoscale variability. Future work should continue to disentangle the overlapping signatures of waves, eddies, and boundary-layer processes, using structure functions as a unifying paradigm across platforms.

This study represents one contribution of the S-MODE experiment, which involved a myriad of additional measurements (Farrar et al. 2025) not explicitly analyzed in this study. Future work can analyze the more completely integrated S-MODE dataset to further contextualize and improve upon the results presented here.

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Data availability statement. The solutions are produced with ROMS, code publicly available on https://github.com/CESR-lab/ucla-roms.git

The SMODE ROMS output and configuration files, the wave gliders, the saildrones and the DopplerScatt datasets are publicly available through the NASA Physical Oceanography Distributed Active Archive Center (PO.DAAC). The buoy data can be accessed through their real-time data repository at https://www.ndbc.noaa.gov (NOAA National Data Buoy Center).

APPENDIX A

Deriving the connection between longitudinal and transverse structure functions and correlation functions

A1. Simplifying $S_{||}(\mathbf{r})$

We start by expanding the dot product and using the definition of $\mathbf{r} \equiv r(\cos\theta, \sin\theta)$:

$$\begin{aligned} S_{||}(\mathbf{r}) &= \frac{1}{r^2}\langle([\mathbf{u}(\mathbf{x}+\mathbf{r}) - \mathbf{u}(\mathbf{x})]\cdot\mathbf{r})^2\rangle_{\mathbf{x}} \\ &= \frac{1}{r^2}\langle[\mathbf{u}(\mathbf{x}+\mathbf{r})\cdot\mathbf{r} - \mathbf{u}(\mathbf{x})\cdot\mathbf{r}]^2\rangle_{\mathbf{x}} \\ &= \frac{1}{r^2}\langle[r(u(\mathbf{x}+\mathbf{r})\cos\theta + v(\mathbf{x}+\mathbf{r})\sin\theta) \\ &\quad - r(u(\mathbf{x})\cos\theta + v(\mathbf{x})\sin\theta)]^2\rangle_{\mathbf{x}} \end{aligned}$$

Set,

$$(u', v') \equiv [u(\mathbf{x}+\mathbf{r}), v(\mathbf{x}+\mathbf{r})], \quad (u, v) \equiv [u(\mathbf{x}), v(\mathbf{x})], \tag{A1}$$

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to simplify further:

$$
\begin{aligned}
S_{||}(\mathbf{r}) = \langle &(u'\cos\theta + v'\sin\theta)^2 \\
&- 2(u'\cos\theta + v'\sin\theta)(u\cos\theta + v\sin\theta) \\
&+ (u\cos\theta + v\sin\theta)^2\rangle_{\mathbf{x}} \\
&= \langle u'^2\cos^2\theta + 2u'v'\cos\theta\sin\theta + v'^2\sin^2\theta \\
&\quad - 2(u'u\cos^2\theta + u'v\sin\theta\cos\theta + \\
&\quad v'u\sin\theta\cos\theta + v'v\sin^2\theta) \\
&\quad + u^2\cos^2\theta + 2uv\cos\theta\sin\theta + v^2\sin^2\theta\rangle_{\mathbf{x}}
\end{aligned}
$$

Now, we introduce the correlation functions,

$$
\begin{aligned}
\langle u'u\rangle_{\mathbf{x}} &= R_{uu}(r,\theta)\,, \quad \langle v'v\rangle_{\mathbf{x}} = R_{vv}(r,\theta)\,, \\
\langle u'v\rangle_{\mathbf{x}} &= R_{uv}(r,\theta)\,, \quad \langle uv'\rangle_{\mathbf{x}} = R_{vu}(r,\theta)\,,
\end{aligned}
$$

together with the assumption of homogeneity (as in Eq. (6))

$$
\langle u'^2\rangle_{\mathbf{x}} = \langle u^2\rangle_{\mathbf{x}}\,, \quad \langle v'^2\rangle_{\mathbf{x}} = \langle v^2\rangle_{\mathbf{x}}\,, \quad \langle u'v'\rangle_{\mathbf{x}} = \langle uv\rangle_{\mathbf{x}}\,. \tag{A2}
$$

Substituting these, we obtain:

$$
\begin{aligned}
S_{||}(\mathbf{r}) = 2\langle u^2\rangle_{\mathbf{x}}\cos^2\theta + 2\langle v^2\rangle_{\mathbf{x}}\sin^2\theta + 4\langle uv\rangle_{\mathbf{x}}\cos\theta\sin\theta \\
- 2[R_{uu}\cos^2\theta + (R_{uv} + R_{vu})\sin\theta\cos\theta + R_{vv}\sin^2\theta]
\end{aligned} \tag{A3}
$$

A2. Simplifying $S_{\perp}(\mathbf{r})$

We expand the cross product (in 2D) and use the definition of $\mathbf{r}$:

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$$
\begin{aligned}
S_\perp(\mathbf{r}) &= \frac{1}{r^2}\langle([\mathbf{u}(\mathbf{x}+\mathbf{r})-\mathbf{u}(\mathbf{x})]\times\mathbf{r})^2\rangle_\mathbf{x} \\
&= \frac{1}{r^2}\langle[(u'-u,v'-v)\times r(\cos\theta,\sin\theta)]^2\rangle_\mathbf{x} \\
&= \langle[(u'-u)r\sin\theta-(v'-v)r\cos\theta]^2\rangle_\mathbf{x} \\
&= \langle(u'r\sin\theta-ur\sin\theta-v'r\cos\theta+vr\cos\theta)^2\rangle_\mathbf{x}
\end{aligned}
$$

Expanding and simplifying:

$$
\begin{aligned}
S_\perp(\mathbf{r}) = \langle & u'^2\sin^2\theta-2u'u\sin^2\theta+u^2\sin^2\theta \\
& -2u'v'\sin\theta\cos\theta+2u'v\sin\theta\cos\theta+ \\
& +2uv'\sin\theta\cos\theta-2uv\sin\theta\cos\theta \\
& +v'^2\cos^2\theta-2v'v\cos^2\theta+v^2\cos^2\theta\rangle_\mathbf{x}
\end{aligned}
$$

Using the same correlation functions as before, we get:

$$
\begin{aligned}
S_\perp(\mathbf{r}) = {} & 2\langle u^2\rangle_\mathbf{x}\sin^2\theta+2\langle v^2\rangle_\mathbf{x}\cos^2\theta-4\langle uv\rangle_\mathbf{x}\sin\theta\cos\theta \\
& -2[R_{uu}\sin^2\theta-(R_{uv}+R_{vu})\sin\theta\cos\theta+R_{vv}\cos^2\theta]
\end{aligned} \tag{A4}
$$

we can combine Eqs. (A3) and (A4), to easily recover $S(\mathbf{r}) = \frac{1}{2}(S_\perp(r)+S_\parallel(r))$ through a combination of trignometric identities and cancellations. These expressions are a lot more complex than that of $S(\mathbf{r})$ but some intuition can be obtained by understanding the transformation of the correlation functions along the longitudinal and transverse directions.

APPENDIX B

Asymptotic behavior of second-order SF at small separation scale r

Given a 2D surface velocity field (u,v) in the Cartesian coordinate system (x,y), we expand the second-order velocity structure function (SF noted S_2) for small separation scale r:

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a. Total SF

The Taylor expansion of $\mathbf{u}(\mathbf{x}+\mathbf{r})$ for $\mathbf{r} \to \mathbf{0}$ is

$$\mathbf{u}(\mathbf{x}+\mathbf{r}) \simeq \mathbf{u}(\mathbf{x}) + \mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x}) + \frac{1}{2}(\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x}) + O(r^3)\ , \tag{B1}$$

keeping the second-order. This leads to

$$\begin{aligned} S_2(\mathbf{r}) &\simeq \frac{1}{2}\langle||\mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x}) + \frac{1}{2}(\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})||^2\rangle_{\mathbf{x}} \\ &= \frac{1}{2}\langle(\mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x}))^2 + \frac{1}{4}\left((\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\right)^2 \\ &\quad + \mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x})(\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\rangle_{\mathbf{x}}\ , \end{aligned} \tag{B2}$$

where $\langle\cdot\rangle_{\mathbf{x}}$ is a space average here. In Eq. (B2), the first term is $O(r^2)$, the second term is $O(r^4)$ and the last term is $O(r^3)$. We take the azimuthal integral $1/2\pi\int_0^{2\pi}\cdot d\theta$ to make SF one dimensional. We use $\mathbf{r} = (r_x, r_y) = (r\cos\theta, r\sin\theta)$. Since r is independent of space and time, it can be removed of the averaging operation.

(i) The $O(r^2)$ term:

$$\begin{aligned} 1/2\pi\int_0^{2\pi}\langle(\mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x}))^2\rangle_{\mathbf{x}}d\theta &= \frac{r^2}{2}\langle u_x^2 + v_x^2 + u_y^2 + v_y^2\rangle_{\mathbf{x}} \\ &= \frac{r^2}{4}\langle\zeta^2 + \delta^2 + \alpha^2\rangle_{\mathbf{x}} \end{aligned}$$

So at first order

$$S_2(r) \simeq \frac{r^2}{8}\langle\zeta^2 + \delta^2 + \alpha^2\rangle_{\mathbf{x}}\ . \tag{B3}$$

Which differs from the result of Babiano et al. (1985) with both divergence and strain here.

(ii) The $O(r^3)$ term: Cancels out when taking the azimuthal integration.

(iii) The $O(r^4)$ term:

$$1/2\pi\int_0^{2\pi}\langle\frac{1}{4}((\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x}))^2\rangle_{\mathbf{x}}d\theta$$

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$$= \frac{r^4}{32}\langle 2(u_{xx}u_{yy}+v_{xx}v_{yy})+3(u_{xx}^2+u_{yy}^2+v_{xx}^2+v_{yy}^2)$$

$$+4(u_{xy}^2+v_{xy}^2)\rangle_{\mathbf{x}} ,$$

so

$$S_2(r) \simeq \frac{r^2}{8}\langle \zeta^2+\delta^2+\alpha^2\rangle_{\mathbf{x}} + \frac{r^4}{64}\langle 2(u_{xx}u_{yy}+v_{xx}v_{yy})$$

$$+3(u_{xx}^2+u_{yy}^2+v_{xx}^2+v_{yy}^2)+4(u_{xy}^2+v_{xy}^2)\rangle_{\mathbf{x}} , \tag{B4}$$

when retaining the next order.

1) Longitudinal and transverse SF

The longitudinal SF is

$$S_{\parallel}(\mathbf{r}) = \frac{1}{r^2}\langle([\mathbf{u}(\mathbf{x}+\mathbf{r})-\mathbf{u}(\mathbf{x})]\cdot\mathbf{r})^2\rangle_{\mathbf{x}} . \tag{B5}$$

At first order, it reads

$$S_{\parallel}(r) \simeq \frac{r^2}{8}\langle 3(u_x^2+v_y^2+(u_y+v_x)^2+2u_xv_y)\rangle_{\mathbf{x}} = r^2\langle\frac{\delta^2}{4}+\frac{\alpha^2}{8}\rangle_{\mathbf{x}} . \tag{B6}$$

The transverse SF is

$$S_{\perp}(\mathbf{r}) = \frac{1}{r^2}\langle([\mathbf{u}(\mathbf{x}+\mathbf{r})-\mathbf{u}(\mathbf{x})]\times\mathbf{r})^2\rangle_{\mathbf{x}} . \tag{B7}$$

At first order, it reads

$$S_{\perp}(r) \simeq \frac{r^2}{8}\langle 3(u_y^2+v_x^2+(u_x-v_y)^2+2u_yv_x)\rangle_{\mathbf{x}} = r^2\langle\frac{\zeta^2}{4}+\frac{\alpha^2}{8}\rangle_{\mathbf{x}} . \tag{B8}$$

At first order, this verifies $S_2(r) = 1/2(S_{\perp}(r)+S_{\parallel}(r)) = \frac{r^2}{8}\langle\zeta^2+\delta^2+\alpha^2\rangle_{\mathbf{x}}$.

The ratio is

$$\frac{S_{\perp}}{S_{\parallel}} \sim \langle\frac{\zeta^2+\frac{\alpha^2}{2}}{\delta^2+\frac{\alpha^2}{2}}\rangle_{\mathbf{x}} . \tag{B9}$$

(i) Next orders: $O(r^6)$ terms The next order term of the longitudinal component is given by

$$1/2\pi\int_0^{2\pi}\langle\frac{1}{4}((\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\cdot\mathbf{r})^2\rangle_{\mathbf{x}}d\theta$$

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$$= \frac{1}{4}\frac{r^6}{16}\langle 5(u_{xx}^2 + v_{yy}^2) + 2u_{xx}(u_{yy} + 2v_{xy})$$

$$+2v_{yy}(2u_{xy} + v_{xx}) + (u_{yy} + 2v_{xy})^2 + (2u_{xy} + v_{xx})^2\rangle_{\mathbf{x}} \,, \tag{B10}$$

which is $O(r^6)$. This implies that for the transversal component it is

$$1/2\pi \int_0^{2\pi} \langle \frac{1}{4}((\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\times\mathbf{r})^2\rangle_{\mathbf{x}} d\theta = O(r^6) \,, \tag{B11}$$

as well.

(ii) Next orders: $O(r^4)$ terms In order to satisfy $S_2(r) = 1/2\left(S_\perp(r) + S_\parallel(r)\right)$ at the next order which is $O(r^4)$, we have

$$1/2\pi \int_0^{2\pi} \langle \mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x})(\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\cdot\mathbf{r}\rangle_{\mathbf{x}} d\theta = O(r^4) \,, \tag{B12}$$

for the longitudinal and

$$1/2\pi \int_0^{2\pi} \langle \mathbf{r}\cdot\nabla_x\mathbf{u}(\mathbf{x})(\mathbf{r}\cdot\nabla_x)^2\mathbf{u}(\mathbf{x})\times\mathbf{r}\rangle_{\mathbf{x}} d\theta = O(r^4) \,, \tag{B13}$$

for the transverse.

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